

HEALTHCONNECT EXPERIENCE LAB

# Final Analytics & Decision Support Report

Week 8 — Final Analytics, Dashboard & Business Insights

| Project Information    | Details                                                                           |
|------------------------|-----------------------------------------------------------------------------------|
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| Track                  | Data Analytics                                                                    |
| Positioning            | Performance & Decision Intelligence Analyst                                       |
| Programme              | AnalystLab Africa — Premium Data Analytics Internship                             |
| Project                | HealthConnect Clinic Experience Lab                                               |
| Project objective      | Improving Patient Appointment Attendance and Healthcare Support Using Data and AI |
| Final Analytics status | FINAL   READY FOR SUBMISSION                                                      |

**FINAL EVIDENCE BASIS:** validated Week 7 Analytics outputs carried forward into Week 8. No new analytical storyline or unvalidated KPI has been introduced.

FINAL DATA ANALYTICS DELIVERABLE

## Executive Summary

HealthConnect Clinic seeks to reduce missed appointments, improve patient attendance, make better use of appointment slots, and provide more effective administrative support. The final Data Analytics contribution converts the validated Weeks 5–7 work into a decision-support output focused on where HealthConnect should begin evaluating attendance-support actions and how the result should be monitored.

The dataset contains 5,000 appointment records. Attendance outcome is observable for 4,737 appointments (Attended + No-Show), of which 2,423 are No-Shows, producing a validated Overall No-Show Rate of 51.15%. Reminder Coverage is 72.68% across the full 5,000 appointments.

Booking Lead Time remains the strongest standalone analytical differentiation of No-Show behaviour. Appointments booked 0–30 days in advance show a 39.13% No-Show Rate compared with 63.95% for appointments booked 31–60 days in advance, a +24.82 percentage-point gap. Recorded Previous No-Shows provides complementary prioritisation information: 46.30% for appointments with no recorded previous no-shows versus 57.83% for those with at least one, a +11.53 percentage-point difference.

Reminder Status and Distance to Clinic remain secondary/contextual in the Analytics evidence hierarchy. Reminder Status shows only a weak observational difference and does not establish reminder effectiveness. Distance has a very weak standalone association with No-Show behaviour and does not justify standalone operational targeting.

The final decision-support logic is therefore to prioritise longer-lead appointment groups for further attendance-support evaluation, refine the evaluation population cautiously using Recorded Previous No-Shows, and test reminder, confirmation or other support interventions before wider deployment. The required decision cycle is: Decision → Action → Monitoring KPI → Evaluation → Extend / Adjust / Stop.

**KEY DECISION: prioritise groups for evaluation — do not treat the observed patterns as causal effects or deterministic individual risk predictions.**

## 1. Final Analytics Component

The final Data Analytics component is a validated decision-support solution built around the HealthConnect appointment dataset and the final three-page Power BI dashboard.

| Page            | Final role                          | Interaction rule                |
|-----------------|-------------------------------------|---------------------------------|
| 01 — MONITOR    | Operational performance exploration | Interactive                     |
| 02 — DIAGNOSE   | Validated analytical evidence       | Reference analytical population |
| 03 — PRIORITIZE | Validated decision support          | Reference analytical population |

The dashboard was frozen after visual review, KPI reconciliation, filter-behaviour testing and retesting. Week 7 identified a genuine interpretation risk: interactive subgroup filters could change displayed values while static statements continued to describe the global validated population. The final architecture separates the interactive MONITOR view from the validated DIAGNOSE and PRIORITIZE reference views.

## 2. Final KPIs

| KPI                                | Validated definition / population | Final value |
|------------------------------------|-----------------------------------|-------------|
| Total Appointments                 | Full dataset                      | 5,000       |
| Attendance-Observable Appointments | Attended + No-Show                | 4,737       |
| Overall No-Show Rate               | 2,423 / 4,737                     | 51.15%      |
| Reminder Coverage                  | 3,634 / 5,000                     | 72.68%      |

The denominators are intentionally different. The No-Show Rate uses the attendance-observable population and excludes Cancelled appointments; Reminder Coverage is an operational coverage metric calculated across the full dataset.

## 3. Validated Findings

### 3.1 Primary evidence — Booking Lead Time

| Booking Lead Time | No-Show Rate |
|-------------------|--------------|
| 0–7 days          | 29.47%       |
| 8–14 days         | 35.19%       |
| 15–30 days        | 45.53%       |
| 31–45 days        | 57.03%       |
| 46–60 days        | 71.36%       |

Consolidated comparison: 0–30 days = 39.13%; 31–60 days = 63.95%; gap = +24.82 percentage points. Additional Week 7 validation produced Spearman  $\rho \approx 0.287$  and an odds ratio of approximately 1.42 per additional 10 booking-lead days.

**FINAL CLASSIFICATION: PRIMARY**

### 3.2 Supporting evidence — Recorded Previous No-Shows

Appointments with 0 recorded previous no-shows had a 46.30% No-Show Rate, compared with 57.83% for appointments with at least one recorded previous no-show, a +11.53 percentage-point difference. This variable is used only as complementary group-level prioritisation information.

**FINAL CLASSIFICATION: SUPPORTING / COMPLEMENTARY**

### 3.3 Secondary / contextual evidence

Reminder Status: 54.63% No-Show Rate without a reminder versus 49.86% with a reminder, a +4.78 percentage-point observational difference for no reminder, with a very small effect size ( $\Phi \approx 0.042$ ). This does not establish reminder effectiveness.

Distance to Clinic: Spearman  $\rho \approx 0.055$ , indicating a very weak standalone association with No-Show behaviour. Distance therefore does not support standalone operational targeting in the Analytics workstream.

**FINAL CLASSIFICATION: SECONDARY / CONTEXTUAL**

### 3.4 Robustness and control findings

The Booking Lead Time pattern persisted across Diagnostic (+22.24pp), Follow-up (+28.73pp), General (+23.51pp) and Specialist (+23.16pp) appointment types. However, the Lead Time  $\times$  Appointment Type interaction was not statistically supported ( $p \approx 0.326$ ); no Follow-up-specific rule is therefore justified.

Appointment Day, Appointment Time, Waiting Time, Age Group, Gender, Previous Appointments and Reminder Channel did not provide sufficient standalone decision value to change the final evidence hierarchy.

## 4. Key Visualisations

The final Power BI file is delivered separately as part of the Week 8 package. The key decision visuals are those retained in the validated three-page architecture:

- **MONITOR** — overall appointment outcomes and operational context exploration.
- **DIAGNOSE** — Booking Lead Time and the validated hierarchy of No-Show differentiation.
- **PRIORITIZE** — group-level attention priorities combining the primary Booking Lead Time signal with complementary Recorded Previous No-Shows information.

The presentation should use only the visuals necessary to communicate the decision logic. Exploratory results that did not change the evidence hierarchy are not elevated into final decision visuals.

## 5. Business Insights

| Evidence                                                               | So what?                                                                   | Decision implication                                                                      |
|------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Longer Booking Lead Time has the strongest standalone differentiation. | The observed No-Show Rate rises materially across longer lead-time groups. | Use longer-lead appointments as the primary population for attendance-support evaluation. |
| Recorded Previous No-Shows adds complementary differentiation.         | It can refine group-level prioritisation but is not deterministic.         | Use cautiously to refine the evaluation population, not to label individual patients.     |
| Reminder Status shows weak observational differentiation.              | Current data cannot establish that reminders caused better attendance.     | Test reminder/confirmation strategies before wider deployment.                            |
| Distance is weak standalone Analytics evidence.                        | Distance alone is not a defensible targeting criterion.                    | Do not use Distance as a standalone operational prioritisation rule.                      |

## 6. Actionable Recommendations

### 6.1 PRIORITISE — Longer Booking Lead Times

Use longer-lead appointments as the primary group for further attendance-support evaluation.

Monitoring KPI: No-Show Rate within the evaluated longer-lead population.

### 6.2 REFINE — Recorded Previous No-Shows

Use Recorded Previous No-Shows cautiously as complementary information when refining the population selected for evaluation.

Monitoring KPI: No-Show Rate by Recorded Previous No-Show status within the evaluated population.

### 6.3 TEST & MEASURE — Attendance-Support Strategy

Test reminder, confirmation or other attendance-support interventions before wider deployment. Evaluate measurable improvement relative to a predefined reference.

**VALIDATED EVIDENCE → DECISION → ACTION → MONITORING KPI → EVALUATION → EXTEND / ADJUST / STOP**

## 7. Analytical Limitations and Guardrails

- The appointment record remains the validated unit of analysis; repeated patient identifiers do not support reliable reconstruction of complete longitudinal patient histories.
- Attendance analysis excludes Cancelled appointments; operational coverage metrics may use the full dataset.
- Recorded Previous No-Shows is interpreted as a recorded appointment-level attribute, not a reconstructed longitudinal patient history.
- Distance and Waiting Time contain a small proportion of missing observations.
- Some extreme segments have limited sample sizes and require cautious interpretation.
- The dataset is observational; associations do not establish causal effects.
- Attendance-support intervention effectiveness remains unvalidated.
- The editable Power BI file requires governance so that validated page behaviour and interpretation rules are preserved.

**GUARDRAIL: group-level analytical differentiation ≠ deterministic individual prediction ≠ causal intervention effectiveness.**

## 8. Cross-Track Contribution and Final Integration Status

### 8.1 Data Analytics × Data Science — completed and validated

Data Analytics provided validated findings on Booking Lead Time, Recorded Previous No-Shows, Distance to Clinic and the Lead Time × Appointment Type interaction, together with two explicit modelling test requests.

| Cross-track test             | Data Science result                                                                                                                                                                                             | Final action                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| CT01 — long_lead_followup    | No statistically meaningful incremental predictive value demonstrated. Single-split AUC difference +0.0017; bootstrap 95% CI [-0.0008, 0.0041]; grouped 5-fold CV mean difference +0.0009.                      | Removed from the Week 8 candidate feature set.                                                                                     |
| CT02 — distance_to_clinic_km | Standalone Analytics association was very weak, but grouped 5-fold CV favoured the model with Distance in 5/5 folds; mean AUC difference +0.0043; Top-20%-risk lift 1.44× with Distance vs 1.40× after removal. | Retained in the Data Science model as a modest multivariable contributor; Analytics classification remains Secondary / Contextual. |

This collaboration produced concrete model-feature decisions and preserved a critical interpretation boundary: standalone analytical differentiation, multivariable predictive contribution and causality are different evidence types.

### 8.2 Data Analytics → Project Management — integration attempt documented

The final Analytics findings, decision-support logic and outputs were prepared for Project Management integration, and collaboration was initiated. No Project Management feedback or output was received before finalisation. Consequently, no PM-provided status, decision, priority, risk or readiness conclusion has been inferred or incorporated into this report.

The Data Analytics deliverables were therefore finalised using the validated Week 7 evidence base. The absence of PM feedback is documented as an unresolved final-integration limitation; it does not reopen or invalidate the completed Week 7 Analytics validation.

**STATUS: Analytics component FINAL. Complete HC-POD / PM integration is not claimed as validated without supporting evidence.**

## 9. Decision Traceability

| Business question                                                            | Evidence                                                                                              | Insight                                                                 | Recommendation                                                                  |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Where should HealthConnect begin if it wants to evaluate attendance support? | Booking Lead Time: 39.13% NSR at 0–30 days vs 63.95% at 31–60 days (+24.82pp).                        | Longer lead time is the strongest validated standalone differentiation. | Prioritise longer-lead appointment groups for evaluation.                       |
| Can prioritisation be refined?                                               | Recorded Previous No-Shows: 46.30% with 0 vs 57.83% with ≥1 (+11.53pp).                               | Provides complementary group-level information.                         | Use cautiously to refine the evaluation population.                             |
| Can the current data prove an intervention works?                            | Reminder Status difference is weak and observational; intervention effectiveness has not been tested. | Observed association is not causal effectiveness.                       | Test & Measure before wider deployment; monitor against a predefined reference. |

## 10. Final Readiness Statement

The Data Analytics workstream is finalised for Week 8 submission. Core KPI definitions, analytical findings, evidence hierarchy, dashboard architecture, decision-support logic, recommendations and the Data Analytics × Data Science interpretation were validated during Week 7 and carried forward without introducing a new analytical storyline.

The remaining uncertainty is principally operational: the effectiveness of any attendance-support intervention has not yet been established. Known structural limitations around longitudinal patient history, sparse extreme segments, missing values in selected variables and causal interpretation remain documented.

Final submission QA must preserve consistency across this report, the frozen Power BI dashboard, the Week 7 validation evidence, cross-track evidence and the separate stakeholder presentation materials.

**FINAL DATA ANALYTICS STATUS: READY FOR WEEK 8 SUBMISSION | PM FEEDBACK NOT RECEIVED BEFORE FINALISATION**

## Appendix A — Evidence Base

- WK7 HealthConnect Analytics Testing & Refinement Report / Notebook.
- WK7 HealthConnect Project Summary.
- WK7 HealthConnect Testing & Validation Evidence Pack.
- WK7 HealthConnect HC-POD Cross-Track Testing & Validation Evidence.
- WK7 HealthConnect DA-DS Cross-Track Testing & Validation Evidence.
- WK7 HealthConnect Analytics Dashboard (Power BI).
- HealthConnect Appointment Data.
- Official AnalystLab Africa Week 8 Data Analytics Assignment.
- Week 8 Mentor Session Report.

## Appendix B — Package Structure

This report is the main narrative document of the HealthConnect Final Analytics & Decision Support Package. The final Power BI dashboard and the stakeholder presentation slides/materials are delivered as separate files within the same submission package.